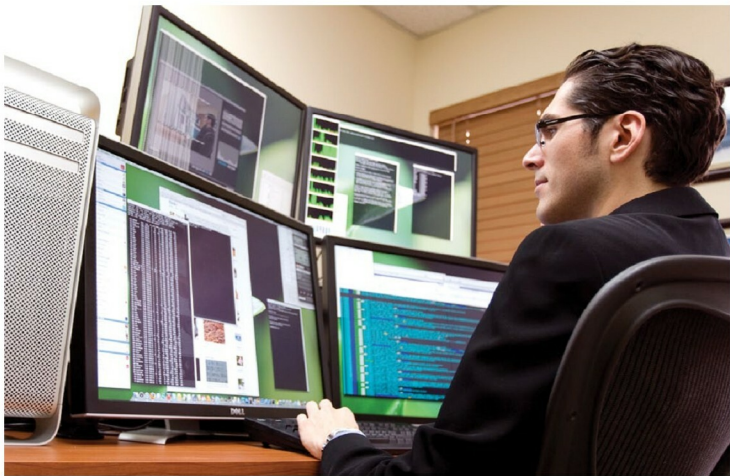


A Detailed Review of Distributed Network Monitoring

Distributed network monitoring involves receiving performance reports about the state of the network from various locations in it, via different types of pollers. This article deals with the concept at great length.



Distributed monitoring involves the use of the underlying distributed infrastructure and communication protocols. Each server or device runs a client that acts as a node in the distributed monitoring network. The node collects metrics, is topology aware, uses gossip protocol for communication and contributes in monitoring other servers. This approach promises robustness and low latency failure detection.

Overlay network monitoring enables distributed Internet applications to detect and recover from path outages and periods of degraded performance, within seconds. This makes it possible to manage distributed systems such as VPN's (virtual private networks) and CDN's (content distribution networks). Accurate loss rate monitoring is critical to building adaptive overlay systems like streaming media frameworks.

In this article, we discuss the various building blocks of monitoring infrastructure, highlight feature capabilities, the significance of 'probe' vs 'router' models and look at new approaches to distributed network monitoring like packet doppler, gossip based monitoring and network tomography.

Network monitoring is the practice of monitoring infrastructure over a computer network to ensure service delivery to meet service level agreement (SLA) requirements and performance metrics for hardware infrastructure, networking equipment and repositories of virtual servers and software infrastructure.

With the convergence of telecommunications infrastructure with IP infrastructure to deliver seamless services over the network, operational practices have seen an overlap of frameworks, practices and protocols.